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Work machine

Abstract

A work machine according to the present invention includes a manual operating tool configured to operate a hydraulic actuator; an operation amount detector configured to detect an operation amount of the manual operating tool; a hydraulic pump configured to generate a hydraulic pressure to be supplied to the hydraulic actuator; an electric pump motor configured to change an output characteristic of the hydraulic pump in response to a rotary force to be given to the hydraulic pump; a motor load detector configured to detect a motor load of the pump motor; and a pump motor controller configured to determine a necessary output characteristic based on the operation amount and the motor load and control the pump motor based on the output characteristic.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATION

(1) This application claims priority to Japanese Patent Application No. 2023-115036 filed Jul. 13, 2023, the disclosure of which is hereby incorporated by reference in its entirety.

BACKGROUND OF THE INVENTION

1. Field of the Invention

(2) The present invention relates to a work machine including a hydraulic actuator and a hydraulic pump driven by an electric pump motor.

2. Description of Related Art

(3) A vehicle according to Japanese Unexamined Patent Application Publication No. 2013-203377 (Patent Literature 1) includes: a hydraulic fixture including an electric hydraulic pump configured to supply a hydraulic pressure to a hydraulic actuator; an operation amount detector configured to detect an operation amount for the hydraulic fixture which operation amount is given by an operating tool; a pump rotation speed calculator configured to receive the operation amount as an input parameter and calculate a pump rotation speed for the electric hydraulic pump which pump rotation speed is necessary for the hydraulic pressure to be supplied to the hydraulic actuator; and a hydraulic pump controller configured to output a control signal to the electric hydraulic pump so

that the electric hydraulic pump is driven at the pump rotation speed calculated by the pump rotation number calculation unit. In a hydraulic pump control in this vehicle, a pump rotation speed for the electric hydraulic pump which pump rotation speed should satisfy a necessary hydraulic pressure is calculated based on the operation amount of the operating tool for the hydraulic fixture, and the electric hydraulic pump is controlled to be driven with the calculated pump rotation speed being taken as a target. Patent Literature 1: Japanese Unexamined Patent Application Publication No. 2013-203377

SUMMARY OF THE INVENTION

(4) In the hydraulic pump control according to Patent Literature 1, the input parameter for the control is only the operation amount for an operator to operate the operating tool, and therefore, in a case where the operator performs an inappropriate operation, a drive control on the hydraulic pump may become inappropriate.

(5) In view of such circumstances, an object of the present invention is to provide a work machine that performs a drive control on a hydraulic pump more appropriately.

(6) A work machine according to the present invention includes: a hydraulic actuator; a manual operating tool configured to operate the hydraulic actuator; an operation amount detector configured to detect an operation amount of the manual operating tool; a hydraulic pump configured to generate a hydraulic pressure to be supplied to the hydraulic actuator; an electric pump motor configured to change an output characteristic of the hydraulic pump in response to a rotary force to be given to the hydraulic pump; a motor load detector configured to detect a motor load of the pump motor; and a pump motor controller configured to determine a necessary output characteristic based on the operation amount and the motor load and control the pump motor based on the output characteristic.

(7) In this configuration, the hydraulic pump is controlled to be driven based on the output characteristic necessary for the hydraulic pump which output characteristic is calculated based on the operation amount of the manual operating tool and the motor load of the pump motor configured to give a rotary force to the hydraulic pump. This avoids inefficient drive that can occur when the hydraulic pump is driven only depending on the operation by an operator. The restriction on the inefficient drive of the hydraulic pump also contributes to energy saving and noise reduction. At the same time, the motor load of the pump motor is also taken into consideration in the drive control of the hydraulic pump, thereby also restraining the hydraulic pump from being excessively driven due to an inappropriate manual operation.

(8) A work machine according to the present invention includes: a hydraulic actuator; a manual operating tool configured to operate the hydraulic actuator; an operation amount detector configured to detect an operation amount of the manual operating tool; a hydraulic pump configured to generate a hydraulic pressure to be supplied to the hydraulic actuator; an electric pump motor configured to change an output characteristic of the hydraulic pump in response to a rotary force to be given to the hydraulic pump; a workload estimator configured to estimate a workload of the hydraulic actuator based on the operation amount; and a pump motor controller configured to determine the output characteristic based on the operation amount and the workload and control the pump motor based on the output characteristic.

(9) In this configuration, the hydraulic pump is controlled to be driven based on the output characteristic of the hydraulic pump which output characteristic is calculated based on the operation amount of the manual operating tool and the motor load of the pump motor configured to give a rotary force to the hydraulic pump. This avoids inefficient drive that can occur when the hydraulic pump is driven only depending on the operation by an operator. The restriction on the inefficient drive of the hydraulic pump also contributes to energy saving and noise reduction. Furthermore, the workload (the workload also includes a workload of a work device operated by the hydraulic actuator) of the hydraulic actuator that operates in response to the operation amount of the manual operating tool is also taken into consideration in the drive control of the hydraulic

pump, so that the hydraulic pressure of an appropriate output characteristic necessary for the operation of the hydraulic actuator is generated, thereby achieving an efficient drive control of the hydraulic pump.

(10) In the present invention, it is also suggested that the output characteristic is either a discharge amount or a discharge pressure, or both the discharge amount and the discharge pressure. A movement required for the work device operated by the hydraulic actuator varies depending on the type of the work device. Accordingly, as the output characteristic required for the hydraulic pump, the selection of whether the discharge amount is given priority to, the discharge pressure is given priority to, or both of them are given priority to is performed appropriately based on a movement required for a work device to be used, so that a more appropriate drive control for the hydraulic pump is enabled.

(11) In the present invention, it is also suggested that the hydraulic actuator is a hydraulic cylinder (a bucket cylinder, a boom cylinder, or the like) configured to lift and lower a bucket or a work device as a lifted and lowered object, and the workload estimator estimates a large workload for a lifting operation to lift the lifted and lowered object and estimates a small workload for a lowering operation to lower the lifted and lowered object. In a case where the lifted and lowered object operated by the hydraulic actuator is a bucket of a bucket loader, a workload at the time when the bucket in a loaded state is lifted against the gravity, a workload at the time when the bucket in a loaded state is lowered along the gravity, a workload at the time when the bucket in an unloaded state is lifted against the gravity, and a workload at the time when the bucket in an unloaded state is lowered along the gravity are different from each other. Such a workload is estimated and the pump motor is controlled in response to the output characteristic determined based on the estimated workload and the operation amount, so that a more appropriate drive control can be performed on the hydraulic pump. Similarly, even in a case where the lifted and lowered object is a work device such as a cultivator that is lifted and lowered by a lifting and lowering cylinder, a workload at the time when the work device is lifted and a workload at the time when the work device is lowered are different from each other. Accordingly, such a workload is estimated and the pump motor is controlled based on the output characteristic determined based on the estimated workload and the operation amount, so that a more appropriate drive control can be performed on the hydraulic pump.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

- (1) FIG. 1 is a left side view of a tractor;
- (2) FIG. 2 is a front view of the tractor;
- (3) FIG. 3 is a vertical-section left side view illustrating the arrangement of constituents of a power system;
- (4) FIG. 4 is a functional block diagram of a control system of a pump motor configured to drive a hydraulic pump; and
- (5) FIG. 5 is a flowchart illustrating an exemplary procedure of a pump motor control for the hydraulic pump.

DESCRIPTION OF THE INVENTION

(6) An embodiment to carry out the present invention will be described with reference to the drawings. Note that, in the following description, a direction of an arrow U in FIGS. 1, 2, 3 indicates an “upper side,” a direction of an arrow D indicates a “lower side,” a direction of an arrow F in FIGS. 1, 3 indicates a “front side,” a direction of an arrow B indicates a “rear side,” a direction of an arrow R in FIG. 2 indicates a “right side,” and a direction of an arrow L indicates a “left side,” unless otherwise specified.

(7) With reference to FIGS. **1, 2, 3**, a work machine according to this embodiment will be described. The work machine includes right and left front wheels **1** and right and left rear wheels **2** that are attached to a body **3**. The work machine includes a bonnet **4** provided in a front portion of the body **3**, and a driving section **5** provided in a rear portion of the body **3**. The driving section **5** includes a control lever **6** configured to steer the front wheels **1**, a driver seat **7**, and a ROPS frame **9**. A plurality of manual operating tools **10** for operating a work device provided in the work machine is disposed around the control lever **6**.

(8) The body **3** illustrated in FIGS. **1, 3** is a frame structure. As illustrated in FIG. **3**, the work machine includes a transmission **12** attached to the rear portion of the body **3** and including a continuously variable transmission **11**, and a front axle case **8** attached to the front portion of the body **3**. The front wheels **1** are supported on the front axle case **8**, and the rear wheels **2** are supported on a transmission case **13** storing the transmission **12**.

(9) As illustrated in FIG. **3**, a lifting and lowering link mechanism **14** is connected to a rear portion of the transmission case **13** in such a manner as to be vertically swingable. Various work devices (not illustrated) can be attached to the lifting and lowering link mechanism **14**. When the lifting and lowering link mechanism **14** is operated to swing vertically by a lifting and lowering cylinder **15**, a work device attached thereto is lifted and lowered. A hydraulic pump **61** and a pump motor **62** configured to drive the hydraulic pump **61** are attached to an upper portion of the transmission case **13**. The pump motor **62** is an electric motor. The lifting and lowering cylinder **15** performs the lifting and lowering operation based on a hydraulic oil from the hydraulic pump **61** as a drive source.

(10) As illustrated in FIGS. **1, 2**, a front loader **17** as one of the work devices is attached to the front portion of the body **3**. The front loader **17** includes a base **18**, a boom **19**, a bucket **20**, a hydraulic boom cylinder **21**, a bucket cylinder **22**, and so on. The boom **19** is lifted and lowered by the boom cylinder **21**, and the bucket cylinder **22** causes the bucket **20** to perform a scoop operation and a dump operation. The hydraulic oil from the hydraulic pump **61** is supplied to the boom cylinder **21** and the bucket cylinder **22**. The hydraulic oil from the hydraulic pump **61** is also supplied to a lift cylinder (not illustrated) configured to swing the lift arm **16**.

(11) A hydraulic actuator OA such as the lifting and lowering cylinder **15**, the boom cylinder **21**, or the bucket cylinder **22** works based on the operation of the manual operating tool **10**.

(12) As illustrated in FIGS. **1, 2**, a mower deck **24** is attached to the body **3** in such a manner as to be liftable and lowerable and is disposed between the front wheels **1** and the rear wheels **2**.

(13) The mower deck **24** is configured such that three rotating blades (not illustrated) driven to rotate around an axial center along the up-down direction are disposed inside the mower deck **24** to be arranged along the right-left direction, so that grass mowed by the rotating blades is ejected rightward from a right ejection section **25** of the mower deck **24**.

(14) As illustrated in FIG. **3**, the front portion of the body **3** includes a drive battery **27**, an inverter **28**, an electric motor **29**, a cooling fan **32**, a fan motor **33**, and an electric equipment battery **34**. The drive battery **27** is stored in the bonnet **4**. The inverter **28** and the electric motor **29** are disposed below the drive battery **27**. A lower cover **30** is provided below the inverter **28** and the electric motor **29** in such a manner as to cover the inverter **28** and the electric motor **29**.

(15) As illustrated in FIG. **3**, power from the electric motor **29** is transmitted to the continuously variable transmission **11** via a transmission shaft. The continuously variable transmission **11** is configured as a hydrostatic continuously variable transmission and can continuously shift power to a forward side and a reverse side.

(16) The power shifted by the continuously variable transmission **11** is transmitted to the rear wheels **2** via a sub-transmission (not illustrated) and a rear-wheel differential device (not illustrated) provided inside the transmission **12**. Power branching right before the rear-wheel differential device is transmitted, via a transmission shaft, to a front-wheel differential device (not illustrated) provided inside the front axle case **8** and then transmitted to the front wheels **1** via the

front-wheel differential device.

(17) A front PTO shaft **37** is provided on a lower side of a front portion of the transmission **12** such that the front PTO shaft **37** faces forward, and a rear PTO shaft **38** is provided in a rear portion of the transmission **12** such that the rear PTO shaft **38** faces rearward. In the power transmitted via the transmission shaft, power not shifted by the continuously variable transmission **11** is transmitted to the front PTO shaft **37** and the rear PTO shaft **38**.

(18) Power from the front PTO shaft **37** is transmitted to the mower deck **24** via a transmission shaft (not illustrated). Power from the rear PTO shaft **38** is transmitted to a work device attached to the lifting and lowering link mechanism **14** via a transmission shaft (not illustrated).

(19) The cooling fan **32** and the fan motor **33** are integrated and attached to a front portion of the lower cover **30**. A dust proofing net as a dust proofing tool **31** is attached to a region of the lower cover **30** which region faces the cooling fan **32**. Due to a suction force of the cooling fan **32**, external air as cooler wind is introduced into the lower cover **30** through the dust proofing tool **31**. Due to the introduced cooler wind, the inverter **28** and the electric motor **29** are cooled.

(20) FIG. **4** illustrates a control functional block to control the pump motor **62** configured to provide a rotary force to the hydraulic pump **61** configured to generate a hydraulic pressure as the drive source for the hydraulic actuator OA. Here, a hydraulic cylinder constituting the hydraulic actuator OA, e.g., the lifting and lowering cylinder **15**, the boom cylinder **21**, or the bucket cylinder **22**, works in response to the operation of the manual operating tool **10**. The operation of the manual operating tool **10** is detected by an operation amount sensor **35**. The hydraulic pressure to be supplied to the hydraulic actuator OA is generated by the hydraulic pump **61**. The hydraulic pump **61** receives a rotary force to generate a hydraulic pressure from the pump motor **62**.

(21) A core element in the control functional block illustrated in FIG. **4** is a control unit **100** constituted by a computer base. The control unit **100** in this embodiment includes a pump motor controller **50**, a motor driver **51**, a motor load detector **52**, an operation amount detector **53**, a workload estimator **54**, and a hydraulic circuit controller **70**.

(22) The motor driver **51** gives a driving signal to the pump motor **62** and rotates the pump motor **62**. The motor load detector **52** receives a status signal indicative of a driving state of the motor driver **51** and detects a motor load of the pump motor **62** based on the status signal. Of course, the motor load detector **52** can detect the motor load based on an electrical signal acquired from the pump motor **62**.

(23) The operation amount detector **53** generates operation data indicative of the operation behavior of the manual operating tool **10** based on an operation amount (a signal or data) from the operation amount sensor **35** configured to detect the operation of the manual operating tool **10**. The operation data thus generated is given to a functional section that requires this operation data. The operation data includes at least one of the operation displacement, the operation position, the operation direction, and the operation speed of each hydraulic operating tool constituting the manual operating tool **10**.

(24) The pump motor controller **50** determines an output characteristic required for the hydraulic pump **61** based on the operation amount of the manual operating tool **10**, more specifically, the operation data received from the operation amount detector **53** and the motor load received from motor load detector **52**. Further, the pump motor controller **50** generates a control signal for driving the pump motor **62** so that the hydraulic pump **61** works with the output characteristic. Note that the output characteristic includes either of or both a discharge amount and a discharge pressure of the hydraulic pump **61**.

(25) In this embodiment, the workload estimator **54** configured to estimate a workload required for the hydraulic pump **61** based on the operation data given from the operation amount detector **53** is provided. The workload thus estimated by the workload estimator **54** is given to the pump motor controller **50**. That is, the operation amount detector **53** estimates an action (herein referred to as the workload) of the hydraulic pump **61** which action is predictable from the operation of the

manual operating tool **10** with time. The pump motor controller **50** calculates the output characteristic (the discharge amount: the flow rate and the discharge pressure: the hydraulic pressure) of the hydraulic pump **61** in consideration of the estimated workload and generates a control signal for the pump motor **62**. Hereby, the hydraulic pump **61** is driven appropriately.

(26) In a simple example of the pump motor control in consideration of the workload, in a case where the hydraulic actuator OA is the bucket cylinder **22** or the boom cylinder **21** configured to lift and lower the bucket **20** as an example of a lifted and lowered object, the workload estimator **54** estimates a large workload for a lifting operation (a movement against the gravity) to lift the bucket **20** and estimates a small workload for a lowering operation (a movement helped by the gravity) to lower the bucket **20**. Alternatively, the workload estimator **54** can be configured to estimate a larger workload as the carrying capacity of the bucket **20** is larger or as the lifting and lowering speed of the bucket **20** is larger. When the workload thus estimated is taken into consideration as such, the hydraulic pump **61** can achieve an appropriate output characteristic based on the rotation of the pump motor **62**. Similarly, even in the lifting and lowering operation of the lifting and lowering cylinder **15** configured to lift and lower a work device (a cultivator, a fertilizer spraying device, or the like (not illustrated herein)) as an example of the lifted and lowered object, the workload estimator **54** estimates a large workload for a lifting operation (a movement against the gravity) of the lifting and lowering cylinder **15** and estimates a small workload for a lowering operation (a movement helped by the gravity) of the lifting and lowering cylinder **15**. Hereby, the rotation of the pump motor **62** is controlled appropriately.

(27) An example of the pump motor control in the hydraulic control system will be described with reference to the flowchart of FIG. 5.

(28) When the pump motor control is started, an initial setting process is performed (#01). In the initial setting process, a default control signal for the pump motor **62** is generated such that the hydraulic pump **61** is driven with a default output characteristic, and the motor driver **51** gives a driving signal corresponding to this control signal to the pump motor **62**, so that the pump motor **62** performs initial drive (#02). As a result, the hydraulic pump **61** performs initial drive with the default output characteristic (default output characteristic drive) (#03).

(29) Subsequently, in order to check whether or not it is necessary to change the default output characteristic, two parameters are acquired. Respective parameter acquisition processes of acquiring the two parameters may be performed independently, and only the results may be stored in a memory. One of the parameter acquisition processes is acquisition of a motor load by the motor load detector **52** (#11).

(30) The other one of the parameter acquisition processes is acquisition of an operation amount detected by the operation amount sensor **35** (#21). The operation amount detector **53** generates operation data that can be handled by a functional section of the control unit **100** from the operation amount thus acquired (#22). In this embodiment, whether a pump motor control in consideration of a workload is performed or not is checked (selected) (#23). This may be set by process setting in advance or may be automatically or manually set based on a working state or the like. When the workload is taken into consideration (Yes in #23), the workload is estimated based on the operation data (#24).

(31) Note that the operation data generated in step #22 is also given to the hydraulic circuit controller **70**. The hydraulic circuit controller **70** generates a hydraulic pressure control signal based on the operation data thus given and transmits the hydraulic pressure control signal to a hydraulic circuit (a control valve or the like). Hereby, the hydraulic actuator OA performs an operation corresponding to the operation of the manual operating tool **10**.

(32) In a case of Yes in step #23, the workload estimated in step #24 and the motor load are given to the pump motor controller **50** as a parameter for determination of an output characteristic. In a case of No in step #23, the motor load and the operation data are given to the pump motor controller **50** as a parameter for determination of an output characteristic. The pump motor

controller **50** determines an output characteristic (a necessary output characteristic) required for the hydraulic pump **61** based on the parameter thus given (#31). The output characteristic is determined by use of a function or a lookup table with the parameter such as a motor load, operation data, or a workload being taken as an explanatory variable and the output characteristic being taken as a response variable.

(33) Then, a difference between the output characteristic thus determined and a currently set output characteristic is found to check whether it is necessary to change the output characteristic (#32).

When the difference is small and it is not necessary to change the output characteristic (No in #32), the process returns to step #11 and step #21 to repeat the parameter acquisition process.

(34) When the difference is large and it is necessary to change the output characteristic (Yes in #32), the pump motor **62** is driven such that the hydraulic pump **61** can generate a hydraulic pressure with the output characteristic thus determined. The pump motor controller **50** generates a control signal to drive the pump motor **62** and transmits it to the motor driver **51** (#33). The motor driver **51** gives a driving signal to drive the pump motor **62** based on the received control signal to the pump motor **62** (#34). The rotation of the pump motor **62** in response to the driving signal changes the output characteristic of the hydraulic pump **61** to the output characteristic determined by the pump motor controller **50** (#35).

Alternative Embodiments

(35) (1) In the above embodiment, the hydraulic cylinder is used as the hydraulic actuator OA, but a hydraulic motor may be used, or both of them may be used.

(36) (2) The manual operating tool **10** may be constituted by a plurality of levers and buttons or may be constituted by a multifunctional lever configured such that a single lever is operated multidirectionally. In a case where a plurality of operations occurs at the same time, an integrated operation amount to integrate respective operation amounts of the operations may be calculated, and a parameter for determination of an output characteristic may be generated based on the integrated operation amount.

(37) (3) The functional block diagram illustrated in FIG. 4 is drawn for the purpose of description, and a particular functional section (for example, the motor driver **51** or the hydraulic circuit controller **70**) may be provided outside the control unit **100**. In addition, respective functional sections may be divided or combined freely.

(38) Note that the configurations described in the above embodiment (including the alternative embodiments; the same applies hereinafter) can be applied in combination with configurations of other embodiments as long as no inconsistency occurs. Further, the embodiment disclosed in the present specification is just an example. The embodiment of the present invention is not limited to this, and various modifications can be made within a range that does not deviate from the object of the present invention.

INDUSTRIAL APPLICABILITY

(39) The present invention is not limited to farm equipment such as a tractor and may be also applicable to a civil engineering work machine, a construction work machine, or a stationary work machine.

(40) TABLE-US-00001 Description of the Reference Numerals 10 manual operating tool 11 continuously variable transmission 14 lifting and lowering link mechanism 15 lifting and lowering cylinder 16 lift arm 17 front loader 19 boom 20 bucket 21 boom cylinder 22 bucket cylinder 24 mower deck 27 drive battery 28 inverter 29 electric motor 35 operation amount sensor 50 pump motor controller 51 motor driver 52 motor load detector 53 operation amount detector 54 workload estimator 61 hydraulic pump 62 pump motor 70 hydraulic circuit controller 100 control unit OA hydraulic actuator

Claims

1. A work machine, comprising: a hydraulic actuator; a manual operating tool configured to operate the hydraulic actuator; an operation amount detector configured to detect an operation amount of the manual operating tool; a hydraulic pump configured to generate a hydraulic pressure to be supplied to the hydraulic actuator; an electric pump motor configured to change an output characteristic of the hydraulic pump in response to a rotary force to be given to the hydraulic pump; a motor load detector configured to detect a motor load of the pump motor; a pump motor controller configured to determine the output characteristic based on the operation amount and the motor load and control the pump motor based on the output characteristic; and a workload estimator configured to estimate a workload of the hydraulic actuator based on the operation amount, wherein the pump motor controller determines a necessary output characteristic based on the operation amount and the workload and controls the pump motor based on the output characteristic.
 2. The work machine according to claim 1, wherein: the output characteristic is either a discharge amount or a discharge pressure, or both the discharge amount and the discharge pressure.
 3. The work machine according to claim 1, wherein: the hydraulic actuator is a hydraulic cylinder configured to lift and lower a bucket or a work device as a lifted and lowered object, and the workload estimator estimates a large workload for a lifting operation to lift the lifted and lowered object and estimates a small workload for a lowering operation to lower the lifted and lowered object.
 4. A work machine, comprising: a hydraulic actuator; a manual operating tool configured to operate the hydraulic actuator; an operation amount detector configured to detect an operation amount of the manual operating tool; a hydraulic pump configured to generate a hydraulic pressure to be supplied to the hydraulic actuator; an electric pump motor configured to change an output characteristic of the hydraulic pump in response to a rotary force to be given to the hydraulic pump; a workload estimator configured to estimate a workload of the hydraulic actuator based on the operation amount; and a pump motor controller configured to determine the output characteristic based on the operation amount and the workload and control the pump motor based on the output characteristic.
 5. The work machine according to claim 4, wherein: the output characteristic is either a discharge amount or a discharge pressure, or both the discharge amount and the discharge pressure.
 6. The work machine according to claim 4, wherein: the hydraulic actuator is a hydraulic cylinder configured to lift and lower a bucket or a work device as a lifted and lowered object, and the workload estimator estimates a large workload for a lifting operation to lift the lifted and lowered object and estimates a small workload for a lowering operation to lower the lifted and lowered object.
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